Identification of Unknown Sample “A”

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Introduction

The goal of this analysis was to identify unknown product utilizing the NMR, FTIR, and GC/MS samples that were given. These analyses are quite useful in terms of being 100% certain of the identity of an unknown compound.

Body

NMR

Nuclear Magentic Resonance (NMR) spectroscopy is a technique utilized in chemical analysis. This technique determines the structure and purity of a molecule by making use of the principles of NMR. NMR utilizes magnetic fields to analyze a sample, and these magnetic fields will interact with atomic nuclei. When placed into this magnetic field, these nuclei will absorb this magnetic energy and, in turn, will resonate at a specific frequency, and these interactions can be mapped on a NMR spectra graph. This spectra graph can be analyzed to provide information about the chemical structure and environment of a sample. The chemical shifts and coupling patterns that appear on the NMR spectra graph provide information about the electronic environment around the nuclei, which can allow for the identification of functional groups and determination of molecular structure.

FTIR

Fourier-transform Infrared Spectroscopy (FTIR) is a technique used to obtain the infrared spectrum of absorption or emission of a sample. FTIR works by measuring the absorption of a number of infrared frequencies by a sample. The technique involves passing infrared light waves through a sample and measuring the amount of light absorbed at each frequency. The resulting spectra graph represents molecular absorption and transmission and each spectra is unique depending on the sample being analyzed. When infrared radiation interacts with the sample, it will cause the bonds between atoms to vibrate. Different types of bonds vibrate at

different frequencies. This allows FTIR to be utilized to identify functional groups and molecular structures of the sample.

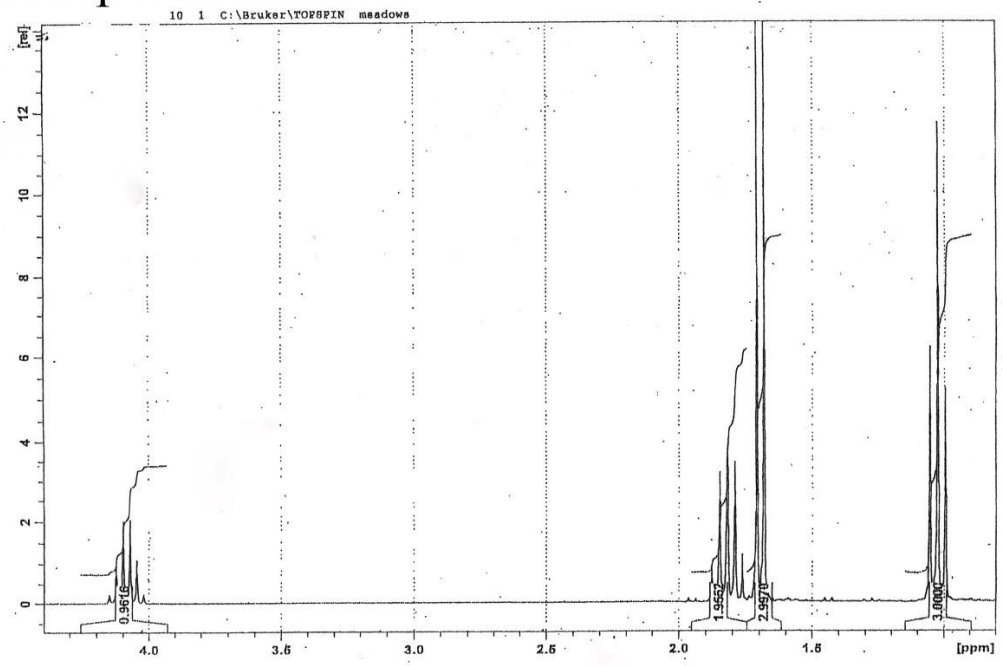
GC/MS

Gas Chromatography-Mass Spectrometry (GC/MS) is an analytical technique used to identify and quantify compounds that are in a complex mixture. The technique combines gas chromatography and mass spectrometry. In gas chromatography, the sample is first converted to vapor form before being injected into a column filled with inert gas. The components of the vaporized sample will then separate based on their interactions with the inert gas. The time taken for each component to travel through the column is that components retention time. In mass spectrometry, the separated compounds from gas chromatography will be put into the mass spectrometer. They are then ionized and fragmented into charged particles before the mass-to-charge ratio and abundance are measured. It is this mass-to-charge ratio and measurement of abundance that provide information about the molecular weight and structure of the compound in the sample. GC/MS relies on two principles: one is the principle that different compounds have different retention times in the gas chromatography column; the second is that different compounds produce unique mass spectra in the mass spectrometer.

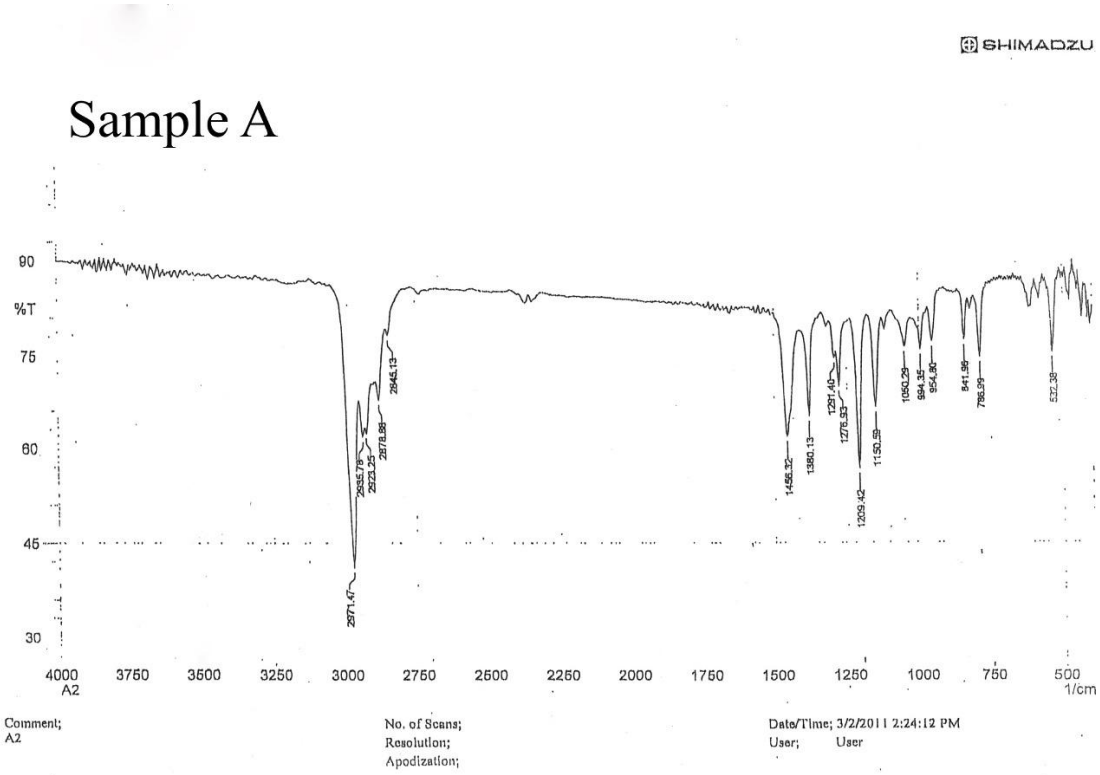
Data

NMR

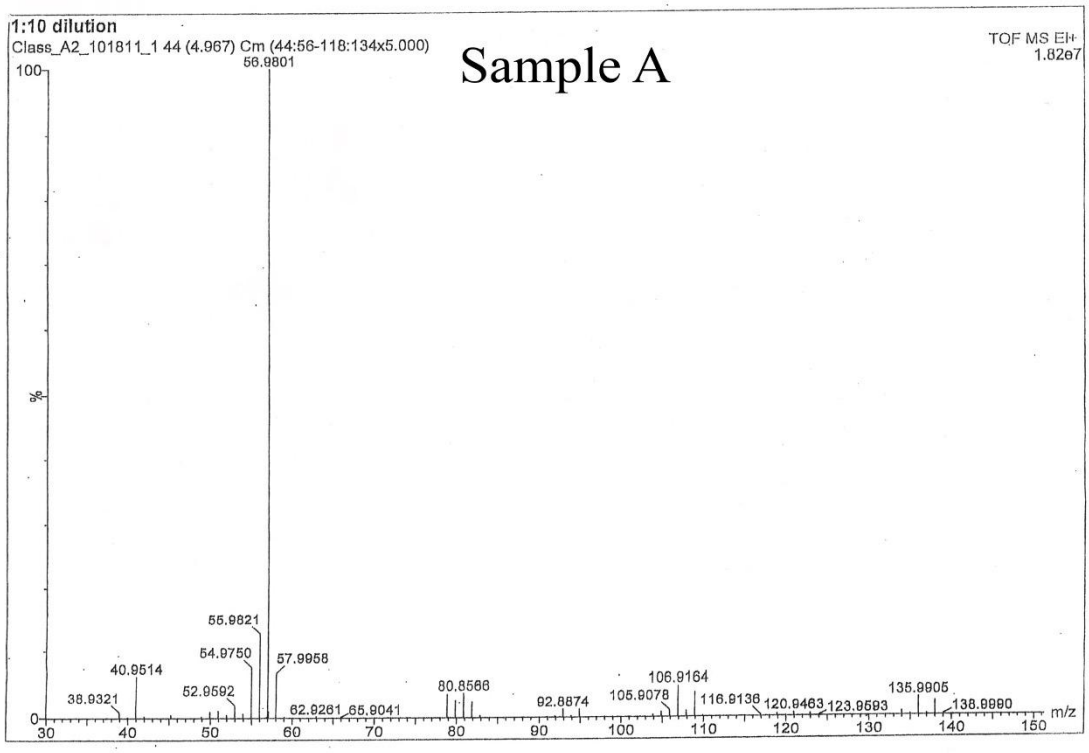
Sample A



FTIR


SHIMADZU
Sample A
90
%T
75
60
45
30
2971.47
2935.79
2923.25
2878.88
2845.13
1456.32
1380.13
1291.40
1276.93
1209.42
1150.58
1050.29
994.35
954.80
841.96
786.09
532.38
4000
3750
3500
3250
3000
2750
2500
2250
2000
1750
1500
1250
1000
750
500
1/cm
A2
Comment;
A2
No. of Scans;
Resolution;
Apodization;
Date/Time; 3/2/2011 2:24:12 PM
User; User

GC/MS


1:10 dilution
Class_A2_101811_1 44 (4.967) Cm (44:56-118:134x5.000)
Sample A
TOF MS EI+
1.82e7
100
%
0
56.9801
55.9821
54.9750
57.9958
40.9514
38.9321
52.9592
62.9261
65.9041
80.8566
92.8874
106.9164
105.9078
116.9136
120.9463
123.9593
135.9905
138.9990
m/z
30
40
50
60
70
80
90
100
110
120
130
140
150

Analysis

NMR

Starting at about 1.05 ppm there is a peak integrated to three. This indicates a methyl group for that peak. At 1.7 ppm there is a peak integrated to two hydrogens. The peak of two hydrogens is split by five. This indicates that it is next to a group of three hydrogens and a group of one hydrogen. At about 1.85 ppm, there is a peak integrated to three with a multiplicity of two. The peak at 4.1 ppm is integrated to one, and based on its shift pattern, the hydrogen is near a strong electron withdrawing group. The shift pattern, multiplicity, and integration shown on the spectra of the molecule indicates a butane with a strong electron-withdrawing group on carbon number two.

FTIR

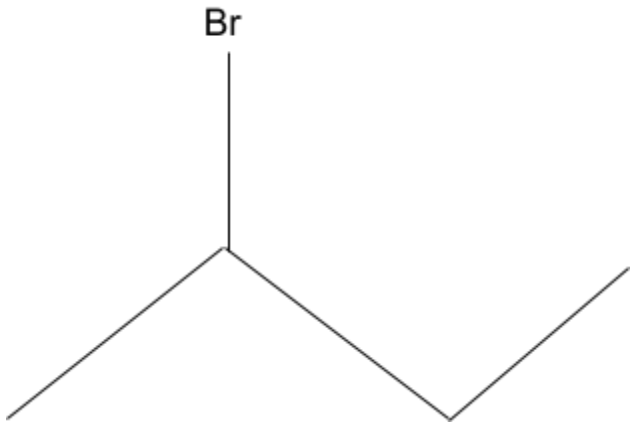
There is a lack of peaks in the 1600-1800 1/cm range indicating no carbonyl groups or alkenes are present in the sample. There is a peak just below 3000 1/cm. This peak signifies an alkane. No broad peaks that would indicate an alcohol group are present. There are also no peaks that would indicate an alkyne.

GC/MS

The strongest peak, which occurs at about 57 m/z, is representative of a butyl group. The two peaks at about 79 and 81 m/z indicate that a bromine is present in this molecule, which is consistent with findings in the NMR analysis. This bromine would act as a an electron-withdrawing group which would help explain the multiplicity that was described in the NMR section. None of the other peaks in this analysis are consistent with isomer data for halogens other than bromine. In addition to this, the data at about 136 m/z indicates the presence of a fully intact 2-bromobutane molecule

Conclusion

Based on the NMR, FTIR, and GC/MS spectra the molecule in sample A is a 2-bromobutane molecule. The NMR peaks are consistent with a molecule that has nine hydrogens and one electron-withdrawing group. The FTIR spectra shows a lack of saturation and has little functionality for determining the unknown other than giving knowledge of the halogen. The GC/MS spectra shows a strong peak at about 57 m/z which indicates a butane group; in addition to this the peaks around 80 m/z indicate the presence of bromine in the sample molecule.



References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.